WORKED SOLUTION

STEP III 2004, Question 1

Pure • Hyperbolic Integrals

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q1

The key idea

Both integrands are a hyperbolic function times something built from the *other* hyperbolic function, and $\sinh x\,\mathrm{d}x$ and $\cosh x\,\mathrm{d}x$ are exactly the differentials of $\cosh x$ and $\sinh x$. So the substitutions $c = \cosh x$ and $s = \sinh x$ turn them into rational integrals: one needing partial fractions, the other an arctangent. The crucial observation for the "hence" is that the two denominators are the same function, since $2\cosh^2 x - 1 = 1 + 2\sinh^2 x$.

Solution

The first integral

Put $c = \cosh x$, so $\mathrm{d}c = \sinh x\,\mathrm{d}x$; the limits $x = 0$ and $x = a$ become $c = 1$ and $c = \cosh a$. Then

$$\int_0^a \frac{\sinh x}{2\cosh^2 x - 1}\,\mathrm{d}x = \int_1^{\cosh a} \frac{\mathrm{d}c}{2c^2 - 1}.$$

The denominator factorises as $(\sqrt{2}\,c - 1)(\sqrt{2}\,c + 1)$, and

$$\frac{1}{(\sqrt{2}\,c - 1)(\sqrt{2}\,c + 1)} = \frac{1}{2}\left(\frac{1}{\sqrt{2}\,c - 1} - \frac{1}{\sqrt{2}\,c + 1}\right),$$

as you can confirm by recombining: the numerator is $\frac{1}{2}((\sqrt{2}c + 1) - (\sqrt{2}c - 1)) = 1$. Each term integrates to a logarithm with an extra factor $1/\sqrt{2}$ from the chain rule. On $c \geq 1$ both $\sqrt{2}c \pm 1$ are positive, so no modulus signs are needed:

$$\begin{aligned}\int_1^{\cosh a} \frac{\mathrm{d}c}{2c^2 - 1} &= \frac{1}{2\sqrt{2}}\Big[\ln(\sqrt{2}\,c - 1) - \ln(\sqrt{2}\,c + 1)\Big]_1^{\cosh a} \\ &= \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}\cosh a - 1}{\sqrt{2}\cosh a + 1}\right) - \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2} - 1}{\sqrt{2} + 1}\right).\end{aligned}$$

Since $-\ln\frac{\sqrt{2}-1}{\sqrt{2}+1} = \ln\frac{\sqrt{2}+1}{\sqrt{2}-1}$, this is exactly the required result.

The second integral

Now put $s = \sinh x$, so $\mathrm{d}s = \cosh x \,\mathrm{d}x$, with limits $s = 0$ and $s = \sinh a$:

$$\int_0^a \frac{\cosh x}{1 + 2\sinh^2 x}\,\mathrm{d}x = \int_0^{\sinh a} \frac{\mathrm{d}s}{1 + (\sqrt{2}\,s)^2} = \frac{1}{\sqrt{2}}\Big[\arctan(\sqrt{2}\,s)\Big]_0^{\sinh a}$$
$$= \frac{1}{\sqrt{2}}\arctan(\sqrt{2}\sinh a).$$

The infinite integral

Ask yourself The two integrands have different-looking denominators. How can we subtract them?

Using $\cosh^2 x = 1 + \sinh^2 x$,

$$2\cosh^2 x - 1 = 2 + 2\sinh^2 x - 1 = 1 + 2\sinh^2 x,$$

so the two denominators are identical, and

$$\int_0^a \frac{\cosh x - \sinh x}{1 + 2\sinh^2 x}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\arctan(\sqrt{2}\sinh a) - \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}\cosh a - 1}{\sqrt{2}\cosh a + 1}\right)$$
$$- \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right).$$

Now let $a \to \infty$. Since $\sinh a \to \infty$, $\arctan(\sqrt{2}\sinh a) \to \frac{\pi}{2}$. Since $\cosh a \to \infty$,

$$\frac{\sqrt{2}\cosh a - 1}{\sqrt{2}\cosh a + 1} = \frac{1 - 1/(\sqrt{2}\cosh a)}{1 + 1/(\sqrt{2}\cosh a)} \to 1,$$

so its logarithm tends to 0, by continuity of ln at 1. All three terms have finite limits, so the infinite integral converges and

$$\int_0^\infty \frac{\cosh x - \sinh x}{1 + 2\sinh^2 x}\,\mathrm{d}x = \frac{\pi}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right),$$

as required.

The final integral

Ask yourself What do $\cosh x - \sinh x$ and $1 + 2\sinh^2 x$ look like in terms of e^x?

With $u = e^x$ we have $\cosh x - \sinh x = e^{-x} = 1/u$, and, using the double-angle formula,

$$1 + 2\sinh^2 x = \cosh 2x = \frac{u^2 + u^{-2}}{2} = \frac{u^4 + 1}{2u^2}.$$

Also $\mathrm{d}u = e^x\,\mathrm{d}x$, so $\mathrm{d}x = \mathrm{d}u/u$, and the limits $x = 0$, $x \to \infty$ become $u = 1$, $u \to \infty$. Hence

$$\int_0^\infty \frac{\cosh x - \sinh x}{1 + 2\sinh^2 x}\,\mathrm{d}x = \int_1^\infty \frac{1}{u}\cdot\frac{2u^2}{1+u^4}\cdot\frac{\mathrm{d}u}{u} = 2\int_1^\infty \frac{\mathrm{d}u}{1+u^4}.$$

Halving the previous result,

$$\int_1^\infty \frac{\mathrm{d}u}{1+u^4} = \frac{\pi}{4\sqrt{2}} - \frac{1}{4\sqrt{2}}\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right).$$

This tidies up: $\dfrac{\sqrt{2}+1}{\sqrt{2}-1} = \dfrac{(\sqrt{2}+1)^2}{(\sqrt{2}-1)(\sqrt{2}+1)} = (\sqrt{2}+1)^2$, so the logarithm is $2\ln(1+\sqrt{2})$ and

$$\int_1^\infty \frac{\mathrm{d}u}{1+u^4} = \frac{\pi}{4\sqrt{2}} - \frac{\ln(1+\sqrt{2})}{2\sqrt{2}} \approx 0.2437.$$

Checks

- At $a = 0$ the first formula gives $\frac{1}{2\sqrt{2}}\left(\ln\frac{\sqrt{2}-1}{\sqrt{2}+1} + \ln\frac{\sqrt{2}+1}{\sqrt{2}-1}\right) = 0$, as it should for an integral over an empty interval.
- The final answer is positive and less than $\int_1^\infty u^{-4}\,\mathrm{d}u = \frac{1}{3}$, which is sensible since $1 + u^4 > u^4$.
- Simpson's rule on $\int_1^\infty \frac{\mathrm{d}u}{1+u^4}$ (after $u = 1/t$ it becomes $\int_0^1 \frac{t^2}{1+t^4}\,\mathrm{d}t$) gives 0.243748, agreeing with the formula to twelve decimal places. The infinite integral in x checks numerically as 0.487495, twice this.

Common slips

- Forgetting the factor $\frac{1}{\sqrt{2}}$ from integrating $\frac{1}{\sqrt{2}c \pm 1}$ or $\frac{1}{1+2s^2}$.
- Not changing the limits when substituting, or leaving the answer in terms of c or s.
- In the last part, forgetting that $\mathrm{d}x = \mathrm{d}u/u$, which gives $2u/(1+u^4)$ and a divergent integral.

Beyond STEP

The final result is part of a classic: $\int_0^\infty \frac{\mathrm{d}u}{1+u^4} = \frac{\pi}{2\sqrt{2}}$, and the substitution $u \mapsto 1/u$ shows the pieces over $[0,1]$ and $[1,\infty)$ are $\int_0^1 \frac{1}{1+t^4}\,\mathrm{d}t$ and $\int_0^1 \frac{t^2}{1+t^4}\,\mathrm{d}t$, whose sum and difference are elementary. In a first-year course you would also meet the slicker route

through contour integration and residues, which evaluates $\int_0^\infty \frac{\mathrm{d}u}{1+u^n} = \frac{\pi/n}{\sin(\pi/n)}$ in one go. Needham's *Visual Complex Analysis* is a beautiful place to learn that idea.

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